Name : Shan
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Sem 2
OOP LAB
Tasks

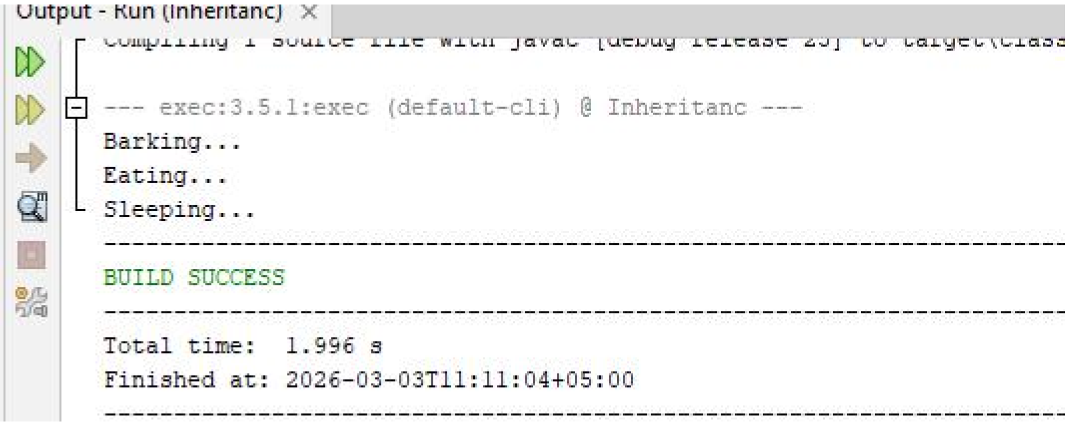
- ## LAB Task 6:

- ## Q1:Write Inheritance Code,Animal,Dog,etc.

Input

```
package com.mycompany.inheritanc;

//AnimalClass

class Animal {
void eat() {
System.out.println("Eating...");
}

void sleep() {
System.out.println("Sleeping...");
}
}

//DogClass

class Dog extends Animal {
void barking() {
System.out.println("Barking...");
}
}

//mainClass

public class Inheritanc  {
public static void main(String[] args) {

Dog d = new Dog();
d.barking();
d.eat();
d.sleep();
}
}
```

Output



✧ Q2:Write Inheritance Code,Parent ,Child,Main etc.

Input

```
package com.mycompany.inheritanc;


// (Parent Class)
class Student {
   void study() {
      System.out.println("Studying basic subjects...");
   }
}

// (Child Class)
class AIStudent extends Student {

   void showDetails() {
      System.out.println("Student Name: Shan");
      System.out.println("ID: A72387");
   }


   void study() {
      System.out.println("Studying Artificial Intelligence and C++ Programming!");
   }
}
```

```
//  MAIN CLASS

public class Inheritanc {
   public static void main(String[] args) {


      AIStudent myStudent = new AIStudent();


      myStudent.showDetails();

      System.out.println("--------------------");

      myStudent.study();
   }
}
```

Output

```
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target\classes

--- exec:3.5.1:exec (default-cli) @ Inheritanc ---
Student Name: Shan
ID: A72387
--------------------
Studying Artificial Intelligence and C++ Programming!
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  1.855 s
```

- **Q3:Write Inheritance Code,Parent ,Child,Main etc.**

Input

```
package com.mycompany.inheritanc;

 //  Parent Class
class Student {
   void study() {
      System.out.println("Studying basic subjects...");
   }
   void giveExam() {
```

```
        System.out.println("Giving normal theory exam...");
    }
}

//Child Class
class AIStudent extends Student {

    void showDetails() {
        System.out.println("Student Name: Shan");
        System.out.println("ID: A72387");
    }

    void study() {
        System.out.println("Studying Artificial Intelligence and C++ Programming!");
    }


    void giveExam() {
        System.out.println("Giving AI Lab Practical Exam on NetBeans!");
    }

    void buildAIModel() {
        System.out.println("Building a smart AI face recognition model...");
    }
}

//  Main Class
public class Inheritanc {
    public static void main(String[] args) {

        AIStudent myStudent = new AIStudent();


        myStudent.showDetails();
        System.out.println("--------------------");
        myStudent.study();


        myStudent.giveExam();
        myStudent.buildAIModel();
    }
}
```

Output

```
Student Name: Shan
ID: A72387
--------------------
Studying Artificial Intelligence and C++ Programming!
Giving AI Lab Practical Exam on NetBeans!
Building a smart AI face recognition model...
------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------
Total time:  1.078 s
Finished at: 2026-03-03T11:33:23+05:00
```